

CAMROSE, AB, Canada

WMO: 712540

Lat: 53.0428N

Lon: 112.8217W

Elev: 2425

StdP: 13.45

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-19.8	-31.8	1.0	-25.6	-25.4	1.5	-19.6	24.4	23.9	22.0	19.1	4.0	270	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	82.8	64.6	79.6	63.2	76.6	61.7	67.9	78.6	65.7	75.8	63.8	73.3	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	99.1	72.7	62.0	91.0	70.0	60.1	84.8	68.0	33.7	78.5	31.9	75.8	30.3	73.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.7	17.2	DB	-34.7	88.9	8.3	3.3	-40.6	91.3	-45.5	93.2	-50.1	95.1	-56.1	97.5
			WB	-34.6	72.1	7.9	2.4	-40.3	73.8	-45.0	75.2	-49.4	76.6	-55.1	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	10.0	12.8	22.7	38.5	50.3	57.6	62.1	59.8	51.1	38.6	23.1	12.2
	DBStd	21.74	16.13	14.06	14.30	9.36	8.00	5.62	5.35	5.84	7.79	8.53	12.41	14.43
	HDD50	6006	1239	1040	846	355	95	8	1	4	79	362	806	1171
	HDD65	10384	1704	1460	1310	794	456	230	123	177	419	819	1256	1636
	CDD50	1152	0	0	0	10	105	235	374	308	112	8	0	0
	CDD65	58	0	0	0	0	2	8	31	16	1	0	0	0
	CDH74	1253	0	0	0	2	98	163	478	381	128	3	0	0
	CDH80	254	0	0	0	0	14	24	110	80	26	0	0	0
Wind	WSAvg	7.7	7.0	6.9	7.9	9.2	9.3	8.4	7.5	6.9	7.6	8.0	7.2	6.9
Precipitation	PrecAvg	19.6	1.1	0.8	1.1	0.8	1.9	3.4	3.3	2.8	1.8	0.7	0.7	1.0
	PrecMax	30.6	2.7	1.9	3.4	2.7	5.7	8.5	6.7	6.7	5.6	2.1	1.8	2.4
	PrecMin	13.7	0.1	0.0	0.1	0.0	0.4	0.4	1.0	0.6	0.1	0.0	0.0	0.3
	PrecStd	3.8	0.7	0.5	0.9	0.6	1.2	2.0	1.6	1.8	1.3	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	44.9	54.8	71.8	82.0	83.2	87.9	86.7	83.8	71.0	55.0	45.3
		MCWB	37.3	37.8	44.0	51.8	57.9	63.3	68.0	65.7	62.2	54.8	44.4	38.3
	2%	DB	39.9	40.0	48.2	66.0	76.4	78.5	82.9	81.9	77.5	63.2	47.2	40.5
		MCWB	35.5	34.9	39.8	49.5	55.8	60.7	67.1	64.7	59.2	50.8	40.0	35.4
	5%	DB	36.3	36.3	43.9	60.0	72.0	74.9	79.3	78.2	72.5	57.5	42.5	36.4
		MCWB	33.1	32.3	37.7	45.9	54.0	59.4	65.3	63.7	57.4	47.0	37.1	32.4
	10%	DB	32.4	32.8	40.0	55.0	67.4	71.1	75.8	74.7	67.2	52.8	38.4	32.5
		MCWB	29.7	30.0	35.4	43.5	51.6	57.9	63.7	62.6	55.1	44.3	34.5	29.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	38.3	44.7	53.1	60.2	66.7	72.1	70.2	63.4	55.6	45.1	39.2
		MCDB	42.4	43.9	52.8	69.7	73.7	78.0	81.6	79.2	78.5	69.1	53.7	44.1
	2%	WB	35.8	35.3	40.7	50.1	57.4	63.6	69.0	67.2	60.5	51.2	40.7	35.6
		MCDB	39.5	39.2	46.8	63.9	72.9	73.5	79.3	77.7	74.5	61.3	46.2	39.9
	5%	WB	33.2	32.7	38.0	47.1	55.3	61.4	66.6	65.1	58.2	48.0	37.6	32.7
		MCDB	36.3	36.0	43.1	58.4	69.4	71.0	76.9	75.3	70.7	55.9	41.7	36.2
	10%	WB	29.6	29.8	35.7	44.4	53.1	59.3	64.7	63.2	55.7	45.2	34.9	29.6
		MCDB	32.2	32.7	39.7	53.6	65.4	68.4	73.9	73.2	65.9	51.7	37.9	32.5
Mean Daily Temperature Range	5% DB	MDBR	19.0	21.2	19.5	21.4	23.8	21.1	22.2	23.8	24.0	21.3	17.4	18.2
		MCDBR	21.4	25.0	23.4	30.9	31.4	27.9	28.4	31.3	34.7	30.0	21.9	20.4
		MCWBR	17.6	20.3	16.1	16.6	14.3	13.0	14.4	16.1	17.7	18.0	15.4	15.9
	5% WB	MCDBR	21.2	24.1	22.3	29.0	28.5	23.5	25.7	28.1	31.6	28.4	20.9	20.2
		MCWBR	17.8	19.7	15.8	16.2	13.5	12.4	14.0	15.6	16.8	17.6	15.0	16.3
Clear-Sky Solar Irradiance	taub	0.261	0.260	0.302	0.328	0.345	0.356	0.355	0.365	0.310	0.280	0.253	0.246	
	taud	2.298	2.385	2.362	2.372	2.377	2.369	2.405	2.344	2.478	2.537	2.452	2.289	
	Ebn at Noon	233	276	283	288	287	284	282	272	278	266	241	216	
	Edh at Noon	19	23	30	34	36	36	35	35	27	20	16	16	
All-Sky Solar Radiation	RadAvg	326	658	1129	1490	1822	1921	1987	1630	1128	660	325	239	
	RadStd	34	53	85	142	124	123	99	112	119	77	36	23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (87 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page